

Recall:

$$f: \mathbb{F}^n \rightarrow \mathbb{F}^m$$

$$A \in \mathbb{F}^{m \times n} \text{ (given)}$$

$$f(x) = Ax$$

$$R(A) = \left\{ y \in \mathbb{F}^m, \text{ s.t. } y = Ax \text{ for some } x \in \mathbb{F}^n \right\}$$

$R(A)$ is a subspace

$$\dim(R(A)) = \text{rank}(A)$$

(Definition of Rank)

Linear Map

Let U, V be vector spaces defined over the same field $\mathbb{F}$.

A map $f: U \rightarrow V$ is said to be linear

if $f(u+v) = f(u) + f(v)$, $u, v \in U$

addition over U *Addition over V*

and $f(\alpha \cdot v) = \alpha \cdot f(v)$, $\alpha \in \mathbb{F}, v \in U$

scalar mult. of U *scalar mult. of V*

$$\textcircled{1} \quad \mathcal{U} = \mathcal{V} = \mathbb{R}$$

$$f(x) = \alpha \cdot x \longrightarrow \text{Linear Map}$$

$$f(x) = \alpha x + \beta \longrightarrow \text{Not a linear map}$$

A linear map must satisfy

$$f(0) = 0 \rightarrow 0 \in \mathcal{V}$$

$\downarrow$
 $0 \in \mathcal{U}$

$$\textcircled{2} \quad \mathcal{U} = \mathbb{F}^n, \quad \mathcal{V} = \mathbb{F}^m$$

$$f(x) = A \cdot x, \quad A \in \mathbb{F}^{m \times n}$$

Verify f is a linear Map.

Let U and V be two finite dim. vector spaces over the same field F .

$$\dim(U) = n, \quad \dim(V) = m$$

Let $T: U \rightarrow V$ be a linear map
Then there exists a matrix representation of T .

Let $\{u_1, \dots, u_n\}$ be a basis of U ,
let $\{v_1, \dots, v_m\}$ " " " " V .

Any vector $x \in U$ satisfies
$$x = \sum_{i=1}^n \alpha_i u_i \text{ for some } \alpha_1, \dots, \alpha_n$$

Similarly $T(x) = \sum_{i=1}^n \beta_i v_i$ for $\beta_1, \dots, \beta_m$

We will show that there exists a matrix

$C \in F^{m \times n}$ s.t.

$$\begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix} = C \cdot \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

Proof:

$$x = \sum_{i=1}^n \alpha_i u_i, \quad x \in U$$

$$\begin{aligned} T(x) &= T\left(\sum_{i=1}^n \alpha_i u_i\right) \quad \text{Linearity of } T \\ &= \sum_{i=1}^n \alpha_i T(u_i) \end{aligned}$$

$$T(u_i) \in \mathcal{V}$$

$$T(u_i) = \sum_{j=1}^m c_{ij} v_j, \quad i=1, 2, \dots, n$$

$$\begin{aligned} T(x) &= \sum_{i=1}^n \alpha_i T(u_i) = \sum_{i=1}^n \alpha_i \sum_{j=1}^m c_{ij} v_j \\ &= \sum_{j=1}^m v_j \underbrace{\sum_{i=1}^n c_{ij} \alpha_i}_{\gamma_j} \end{aligned}$$

$$= \sum_{j=1}^m \gamma_j v_j$$

$$T(x) = \sum_{j=1}^m \gamma_j v_j$$

$\Rightarrow \beta_j = \delta_j$ (Unique Representation)

$$\Rightarrow \underbrace{\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}}_{\beta} = \underbrace{\begin{bmatrix} c_{11} & c_{21} & c_{31} & \dots & c_{n1} \\ c_{12} & c_{22} & c_{32} & \dots & c_{n2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_{1m} & \dots & \dots & \dots & c_{nm} \end{bmatrix}}_C \underbrace{\begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}}_{\alpha}$$

$$\Rightarrow \boxed{\beta = C\alpha}$$

If U and V are finite dimensional vector spaces and $T: U \rightarrow V$ is a linear map, then there is a matrix representation for T .

In particular, the matrix is of size $m \times n$ where $n = \dim(U)$, $m = \dim(V)$

Null space/ kernel of a map $T: U \rightarrow V$

$$N(T) = \{x \in U, \text{ s.t. } T(x) = 0\}$$

$$R(T) = \{y \in V, \text{ s.t. there exists } x \in U \\ y = T(x)\}$$

[Verify $N(T)$, $R(T)$ are subspaces if T is linear]

RANK-NULLITY THEOREM

Suppose U is a finite dimensional vector space. Let $T: U \rightarrow V$ (V is a v.s.) be a linear map.

Then

$$\dim(R(T)) + \dim(N(T)) = \dim(U)$$

Proof:

$N(T)$ is a subspace of U

U is finite dimensional $(\because T \text{ is linear})$

Let $\dim(U) = n$

Let $\dim(N(T)) = k$, $N(T) \subseteq U$

There exists a basis $b_1, \dots, b_k$ of $N(T)$.

There exist vectors $b_{k+1}, b_{k+2}, \dots, b_n$ in $\mathcal{U}$ such that

$\left\{ \underbrace{b_1, \dots, b_k}_{\text{basis of } N(T)}, b_{k+1}, b_{k+2}, \dots, b_n \right\}$ is a basis of $\mathcal{U}$.

Consider any $x \in \mathcal{U}$

there exist scalars

$\alpha_1, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_n$

$$x = \sum_{i=1}^n \alpha_i b_i$$

$$\begin{aligned} T(x) &= T\left(\sum_{i=1}^n \alpha_i b_i\right) = \sum_{i=1}^n \alpha_i T(b_i) \\ &= \sum_{i=k+1}^n \alpha_i T(b_i) \end{aligned}$$

$$T(b_{k+1}) \in R(T), \quad \dots, \quad T(b_n) \in R(T)$$

Any vector in $R(T)$ is a linear combin.
of $T(b_{k+1}), \dots, T(b_n)$

If S_1, S_2 are subspaces, S_2 is finite dim.,

$$S_1 \subseteq S_2$$

$$\dim(S_1) \leq \dim(S_2)$$

$$\Rightarrow R(T) = \text{span} \{ T(b_{k+1}), \dots, T(b_n) \}$$

We will show that $T(b_{k+1}), \dots, T(b_n)$ are lin. indep.

Suppose $\sum_{i=k+1}^n \alpha_i T(b_i) = 0$ for some $\alpha_{k+1}, \alpha_{k+2}, \dots, \alpha_n$

linearity
of T

$$\Rightarrow T\left(\sum_{i=k+1}^n \alpha_i b_i\right) = 0$$

$$\Rightarrow \sum_{i=k+1}^n \alpha_i b_i \in N(T)$$

$$\Rightarrow \sum_{i=k+1}^n \alpha_i b_i = \sum_{i=1}^k \gamma_i b_i \quad \left(\because b_1, \dots, b_k \text{ is a basis for } N(T) \right)$$

$$\Rightarrow \gamma_i = 0, \quad i=1, \dots, k$$

$$\Rightarrow \alpha_i = 0, \quad i=k+1, \dots, n$$

$$\Rightarrow \{ T(b_{k+1}), \dots, T(b_n) \} \text{ is a basis for } R(T)$$

$$\dim(R(T)) = n - k$$

$$\underbrace{k + \dim(R(T))}_{\dim(N(T))} = n$$

Rank

Consider $A \in \mathbb{F}^{m \times n}$

$$\dim(R(A)) = \text{rank}(A)$$

$$A = [a_1 \ a_2 \ \dots \ a_n], \ a_i \in \mathbb{F}^m$$

$$R(A) = \text{span}\{a_1, \dots, a_n\}.$$

$R(A)$ is a subspace of $\mathbb{F}^m$

Properties of Rank

$$\textcircled{1} \dim(N(A)) = n - \text{rank}(A)$$

(Rank-Nullity Theorem)

$$\textcircled{2} \text{rank}(A) \leq n \quad \left(\begin{array}{ccc} \text{"} & \text{"} & \text{"} \end{array} \right)$$

$$\textcircled{3} \text{rank}(A) \leq m$$

($R(A)$ is a subspace of $\mathbb{F}^m$)

$$\text{rank}(A) \leq \min(m, n)$$

If $\text{rank}(A) = \min(m, n)$, we call it a full (row or column) rank matrix

If $\text{rank}(A) = n$, $\Rightarrow a_1, \dots, a_n$ are lin. indep.

$$\textcircled{4} \quad \text{rank}(A) = \text{rank}(A^T) \quad \left(\begin{array}{l} R(A) \subseteq \mathbb{F}^m \\ R(A^T) \subseteq \mathbb{F}^n \end{array} \right)$$

$$[A^T]_{ij} = A_{ji}$$

Let $\text{rank}(A) = r$

$\Rightarrow$ There exists a basis $\{b_1, \dots, b_r\}$ of $R(A)$.

$$a_1 = \sum_{i=1}^r c_{1i} b_i,$$

$$a_2 = \sum_{i=1}^r c_{2i} b_i$$

$\vdots$

$$a_n = \sum_{i=1}^r c_{ni} b_i$$

$$[a_1 \dots a_n] = [b_1 \ b_2 \dots b_p] \begin{bmatrix} c_{11} & \dots & c_{n1} \\ c_{12} & & \vdots \\ \vdots & & \vdots \\ c_{1r} & \dots & c_{nr} \end{bmatrix}$$

Each row of A is a linear combination of " r " rows of C .

$$\Rightarrow R(A^T) \subseteq \text{row-space of } C$$

$$\dim(R(A^T)) \leq r$$

$$\Rightarrow \boxed{\text{rank}(A^T) \leq r = \text{rank}(A)}$$

Repeat same argument by substituting A^T for A

$$\Rightarrow \text{rank}(A) \leq \text{rank}(A^T)$$

(5) If S_1, S_2 are subspaces

$$\dim(S_1 + S_2) = \dim(S_1) + \dim(S_2) - \dim(S_1 \cap S_2)$$

Proof: Discussion

(6) Given $A \in \mathbb{F}^{m \times n}$, there exists $B \in \mathbb{F}^{n \times m}$ such that $AB = I$ if and only if $\text{rank}(A) = m$

[Such a B will be called a right inverse of A .] Denote $B = [b_1, \dots, b_m]$

Proof: $AB = I = [e_1 \ e_2 \ \dots \ e_m]$

$$\Leftrightarrow \{e_1, \dots, e_m\} \in R(A)$$

(Why? If $AB = I$, $e_i = Ab_i \Rightarrow e_i \in R(A)$, $i=1, 2, \dots, m$.
Conversely if $e_1, \dots, e_m \in R(A)$, then

there exists b_i s.t. $Ab_i = e_i$)

Now $\{e_1, \dots, e_m\} \in R(A)$

$$\Leftrightarrow \dim(R(A)) = m$$

Why? Suppose $\{e_1, \dots, e_m\} \in R(A)$. We know that $\{e_1, \dots, e_m\}$ is a basis of $\mathbb{F}^m$ and $R(A)$ is a subspace of $\mathbb{F}^m$. Therefore, $R(A) \subseteq \text{span}\{e_1, \dots, e_m\}$. Since $e_1, \dots, e_m \in R(A)$, then we also have $\text{span}\{e_1, \dots, e_m\} \subseteq R(A)$, implying

that $R(A) = \text{span}\{e_1, \dots, e_m\}$.

Since $e_1, \dots, e_m$ are also LI, $\{e_1, \dots, e_m\}$ is a basis for $R(A) \Rightarrow \dim(R(A)) = m$.

Conversely, if $\dim(R(A)) = m$, since $R(A)$ is a subspace of $\mathbb{F}^m$, we have $R(A) = \mathbb{F}^m \Rightarrow e_1, \dots, e_m \in R(A)$

⑦ Given $A \in \mathbb{F}^{m \times n}$, there exists

$C \in \mathbb{F}^{n \times m}$ such that

$$CA = I$$

if and only if $\text{rank}(A) = n$.

Proof: Apply the proof of (6) on A^T and use the fact that $\text{rank}(A) = \text{rank}(A^T)$

LEFT AND RIGHT INVERSES

AND $Ax = b$

$A \in \mathbb{F}^{m \times n}$ has a left inverse if and only if the following equivalent statements are true!

- (i) there exists $C \in \mathbb{F}^{n \times m}$, $CA = I_{n \times n}$
- (ii) $\text{rank}(A) = n$

C is called a left inverse of A .

If $\text{rank}(A) = n$ and $m > n$, there are multiple left inverses of A .

$A \in \mathbb{F}^{m \times n}$ has a right inverse if and only if the following equivalent statements are true!

- (i) there exists $B \in \mathbb{F}^{n \times m}$, $AB = I_{m \times m}$
- (ii) $\text{rank}(A) = m$

B is called a "right inverse" of A .

If $\text{rank}(A) = m$ and $n > m$, there are multiple right inverses of A .

IMPLICATIONS FOR SOLUTION OF $Ax=b$

Given $b \in \mathbb{F}^m$, $A \in \mathbb{F}^{m \times n}$, solve for x :

$$b = Ax$$

Case I: Suppose A has a "left inverse".

$$\Leftrightarrow \text{rank}(A) = n$$

Case Ia: $b \notin R(A) \Rightarrow$ no solution

Case Ib: $b \in R(A) \Rightarrow$ at least one solution exists

In this case solution is unique. WHY?

Let x_1, x_2 be two solutions of $b = Ax$.

$$\Rightarrow b = Ax_1 = Ax_2 \Rightarrow x_1 - x_2 \in N(A)$$

$$\text{Since } \text{rank}(A) = n \Rightarrow N(A) = \{0\} \Rightarrow x_1 - x_2 = 0$$

How to find the solution?

Let C be a left inverse of A .

then the unique solution is $\boxed{x^* = Cb}$

Case II: Suppose A has a right inverse.

$$\Leftrightarrow \text{rank}(A) = m$$

In this case, $\dim(R(A)) = \dim(\mathbb{F}^m) = m$

$\Rightarrow R(A) = \mathbb{F}^m \Rightarrow b = Ax$ always has

a solution.

Is the solution unique? Not if $n > m$.

If $n > m$, by rank-nullity theorem

$$\dim(N(A)) = n - \text{rank}(A) = n - m > 0$$

$\Rightarrow$ there exists $v \neq 0$ in $N(A)$

$\therefore$ If x_0 is a solution to $b = Ax$, so is

$$x_1 = x_0 + v \neq x_0$$

If B is a right inverse of A , then

$$\bar{x} = Bb \text{ is a solution}$$

WHY?

$$A(Bb) = ABb = b \quad (\text{proof by verification})$$

Alternately:

Consider Bb

We know $b = Ax_0$ for some x_0 (since solution exists)

$$\Rightarrow Bb = B(Ax_0)$$

$$\Rightarrow ABb = \underbrace{ABA}_{=I}x_0 = Ax_0 = b$$

$\Rightarrow Bb$ is a solution to $b = Ax$

More generally, we can say that if X is either a left or right inverse of A , then.

$$AXA = A$$

INVERTIBLE MATRIX

Suppose A has both left and right inverses
then we must have $m=n=\text{rank}(A)$

Suppose B is a left inverse and C is
a right inverse of A

$$\Rightarrow \underbrace{BAC}_I = C \text{ and } B \underbrace{AC}_I = B$$

$$\Rightarrow B = C$$

Moreover there is exactly one matrix which
is both left and right inverse

Suppose there are two right inverses C_1, C_2

$\Rightarrow AC_1 = AC_2 = I \Rightarrow$ each column of
 $C_1 - C_2$ must be in $N(A) \Rightarrow$ each column of
 $C_1 - C_2$ is zero.

$\Rightarrow$ This unique matrix which is both